Hot Water

# Reliable, Efficient Heat Pump Water Heaters

About 10% of California residents use conventional electric water heating—a technology that's expensive to operate and contributes to peak demand. Heat pump water heaters (HPWHs) are an efficient alternative, consuming less than half as much electricity as their conventional cousins (see "All Pumped Up," *HE* Nov/Dec '02, p. 30). However, products available to date have been expensive to manufacture, install, and maintain.

The Watter$aver heat pump water heater, developed by TIAX LLC in conjunction with manufacturer ECR International, has a shorter installation time than other HPWHs and offers the same life expectancy as a standard electric resistance heater. The Watter$aver is now available as a commercial product from ECR. Laboratory and field tests have confirmed its high performance and reliability (see Table 1) . The Watter$aver represents a significant advance over traditional 50-gallon water heaters.

TIAX has field-tested 20 of its 50-gallon Watter$avers in different California climate zones, installing those units in different areas within the home (garage, basement, or laundry room) and in homes with different numbers of occupants and varying degrees of water hardness. Oak Ridge National Laboratory (ORNL) also tested 10 units in the lab and found the estimated life expectancy for these HPWHs to be about 11 years, whereas standard electric units last between 11 and 14 years (see "Oak Ridge in Hot Water," *HE* Jan/Feb '05, p. 8). The payback on the incremental cost of the Watter$aver compared to a conventional electric unit is about 5 to 8 years, based on California rates of $0.1226/kWh at flat rates.

**Table 1. Watter$aver Energy Savings**

| Performance Factor | Watter-$aver | High-Efficiency Electric Resistance | Standard-Efficiency Electric Resistance |
|---|---|---|---|
| Energy factor | 2.48 | 0.94–0.95 | 0.088 |
| Energy savings per year relative to standard unit | 30%–50% | ~8% | NA |

**Table 2. Reduced Electric Demand of HPWH**

| | Winter Morning Peak (kW) | Winter Evening Peak (kW) |
|---|---|---|
| Electric resistance mode | 2.6 | 1.6 |
| HPWH | 1 | 0.7 |
| Demand Reduction | 62% | 56% |

## *Combined Design and Performance*

The main features of the Watter$aver are embodied in its redesigned components, which include a compressor that operates on 120 volts AC (VAC) power instead of 240 VAC. The low-cost condenser is manufactured with thermal mastic—a puttylike substance with good heat transfer characteristics—instead of solder. The Watter$aver also has a two-speed evaporator fan that provides more flexibility than a single-speed fan at less cost than a variable-speed fan and a backup electric resistance heating element. While the fan results in lower efficiency, the main goal of this project was to develop a market-optimized product, especially focused on lowering the costs. The two-speed evaporator fan allows for more efficient operation than a single-speed fan and less efficient operation than a variable-speed fan, but its lower cost means that it provides more bang for the buck than the variable-speed approach.

Researchers at ORNL found significant decreases in electric demand when they measured demand for six Watter$aver heat pump water heaters for six-week periods in summer and winter and compared the results with measurements for the same water heaters operating in electric-resistance mode (see Table 2). The electric demand of the HPWHs was 56%–62% lower than the electric demand for electric-resistance water heaters.

The Watter$aver leads to energy and money savings as well. Participants in TIAX's HPWH tests showed savings of between 28%–52% on the water-heating portion of their electric bills. Another advantage to the Watter$aver is that the HPWH evaporator removes moisture from the environment, providing dehumidification and cooling in addition to hot water when it is installed in conditioned space.

Surveys of those who participated in the field tests showed that they liked the new design. The most popular features were energy and cost savings, dehumidification capabilities, and the ability of the unit to remove odors, given its dehumidification properties of removing moisture, mildew, and mold. However, compressor noise may be a problem when the HPWH is installed near high-occupancy areas. The preliminary specification called for maximum noise output of 60 dBA. A refrigerator, by comparison, has a noise output of about 40 dBA, a garbage disposal 80 dBA, a washing machine 78 dBA, and a dishwasher 75 dBA.

At its current cost, the residential HPWH is economically viable for California households that use electric water heating and consume more than 64 gallons of hot water per day. Households with four or more occupants typically achieve that level of consumption.

## *High Standards at a Lower Cost*

In California, Title 24 building energy standards encourage gas water heaters in residential new construction. However, in areas where natural gas is not available, the Watter$aver can help builders meet the energy efficiency requirements of Title 24. The Watter$aver is also a viable option for replacing existing electric water heaters.

Consumers can purchase the Watter$aver for $1,500–$1,800 installed; contractors can purchase it for $1,000–$1,200 from distributors. ECR hopes to work with utilities and other organizations to provide incentives for Watter$aver purchase and installation.

If HPWHs are to penetrate the market significantly, manufacturers will have to figure out how to reduce the cost per unit. One solution would be to achieve higher production volumes, but another option would be to reduce installation time and effort. Only one person is needed to install a conventional water heater, but two people are needed to cope with the Watter$aver's height and weight. ECR is investigating several technical improvements to the Watter$aver, including making it lighter and incorporating better controls. Overcoming the cost issue and ensuring that the HPWH is reliable should make for the best possible product.

**—Rachel Reiss Buckley**

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*This article was adapted from a California Energy Commission PIER Technical Brief.*

**For more information:**

Get the full report on the Watter$aver at www.energy.ca.gov/pier/final_project_reports/500-04-018.html.

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